Multivitamins in the Prevention of Cardiovascular Disease in Men: The Physicians' Health Study II Randomized Controlled Trial

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Context Though multivitamins aim to prevent vitamin and mineral deficiency, there is a perception that multivitamins may prevent cardiovascular disease (CVD). Observational studies examining regular multivitamin use have been inconsistently associated with CVD, with no long-term clinical trials of multivitamin use. Objective To determine whether long-term multivitamin supplementation decreases the risk of major cardiovascular events among men. Design The Physicians' Health Study II is a randomized, double-blind, placebo-controlled trial of a common daily multivitamin, that began in 1997 with continued treatment and follow-up through June 1, 2011. Setting and Participants A total of 14,641 male U.S. physicians initially aged ≥50 years (mean [± SD] age; 64.3 [± 9.2] years), including 754 men with a history of CVD at randomization, were enrolled. Intervention Daily multivitamin, as Centrum Silver. Main Outcome Measures The primary cardiovascular outcome was a composite endpoint of major cardiovascular events, including nonfatal myocardial infarction (MI), nonfatal stroke, and fatal CVD. Secondary outcomes included MI and stroke individually. Results During a median (interquartile range) follow-up of 11.2 (10.7 to 13.3) years, there were 1,732 confirmed major cardiovascular events. Compared with placebo, there was no significant effect of a daily multivitamin on major cardiovascular events (active and placebo multivitamin groups, 11.0 and 10.8 events per 1,000 person-years; hazard ratio [HR], 1.01; 95% confidence interval [CI], 0.91–1.10; P=0.91). Further, a daily multivitamin had no effect on total MI (active and placebo multivitamin groups, 3.9 and 4.2 events per 1,000 person-years; HR, 0.93; 95% CI, 0.80–1.09; P=0.39), total stroke (active and placebo multivitamin groups, 4.1 and 3.9 events per 1,000 person-years; HR, 1.06; 95% CI, 0.91–1.23; P=0.48), or cardiovascular mortality (active and placebo multivitamin groups, 5.0 and 5.1 events per 1,000 person-years; HR, 0.95; 95% CI, 0.83–1.09; P=0.47). A daily multivitamin was also not significantly associated with total mortality (HR, 0.94; 95% CI, 0.88–1.02; P=0.13). The effect of a daily multivitamin on major cardiovascular events did not differ between men with or without a baseline history of CVD (P, interaction = 0.62). Conclusions A daily multivitamin did not reduce major cardiovascular events, MI, stroke, and CVD mortality after more than a decade of treatment and follow-up.